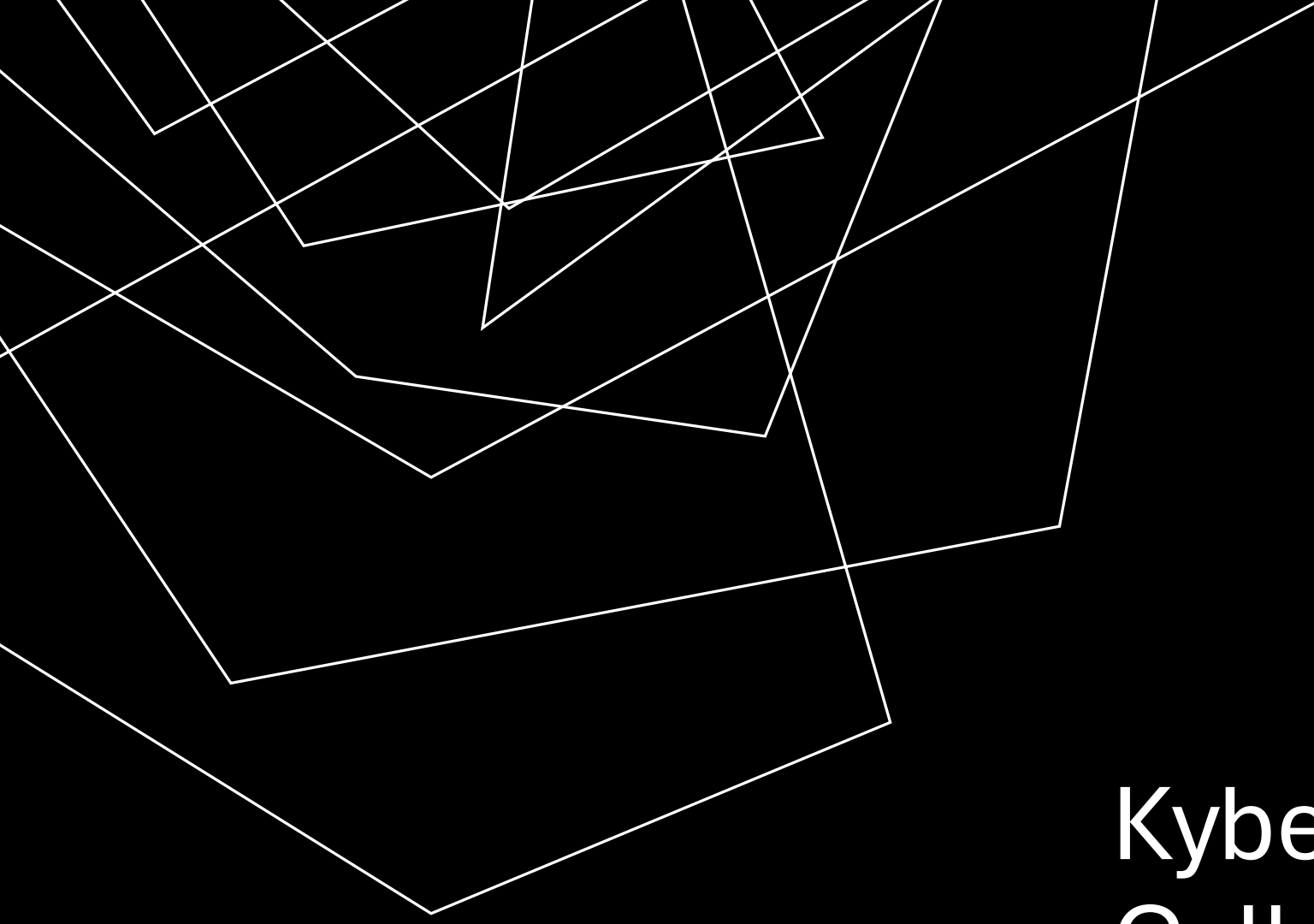




Kyber PQC Chat and Call Room RPI 5

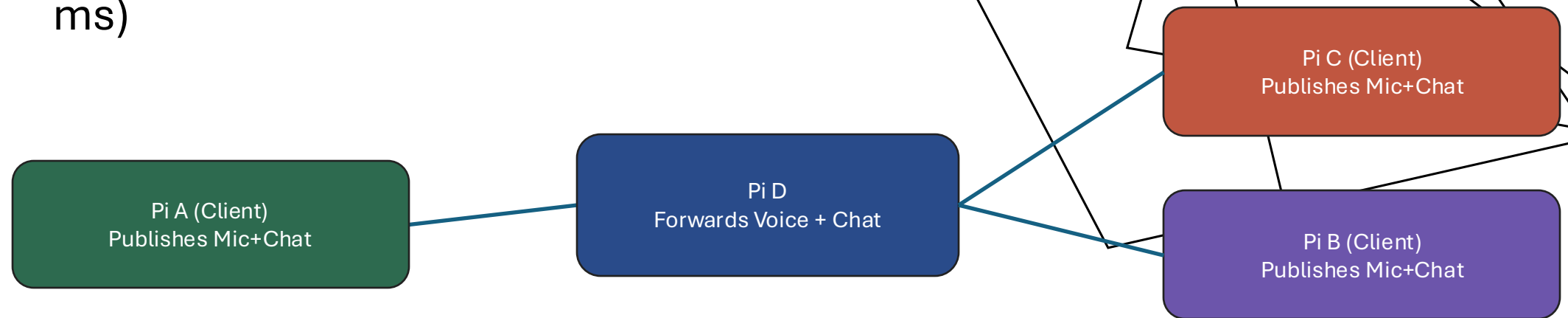
By Aaron Tran, Melvin Contreras, Quinn BELL

Inspiration

- Streaming applications such as Discord, Skype, Messenger, Zoom, Teamspeak, etc.
 - Ex. Discord Channel provides voice and audio calls
 - Allows a Single user to send their voice and video to other members connected to the channel
 - Every user can see each other's voice/video
 - Provide a secure alternative application
 - (Fun Fact this project Idea was conceived over discord)

Secure Pi-to-Pi AV Streaming (1 → 2)

- Architecture: WebRTC mesh (one publisher → two subscribers)
 - Client 1 (Pi A): mic → Opus(audio) → Server-side application → Client 2 (Pi B): Audio Data
 - Signaling: Use Kyber for Key Exchange
 - Security: Kyber key exchange performed for connection to server. TLS is used to send any further data(chat+audio)
 - LAN friendly: On the same Network as the Server, no Server-side connection required
 - Latency target: 80 ms E2E (Opus 20 ms frames, ptime=20, jitter 60–100 ms)



Implementation Plan (Rust, Raspberry Pi 5)

Roles & Responsibilities

- Server (Pi D): Chat + Opus (audio). Handles rooms, RTCP (NACK/PLI), and bitrate guidance.
- Records room participant count as well as displays users connected to server
- Clients (Pis A/B/C): publish mic and chat; subscribe to room streams; UI controls (mute/PTT/video off).

Signaling & Security

- Kyber Key Exchange and TLS forwarding for audio + chat data
- Access control lists per room; join/leave events for presence.

Networking & QoS

- LAN-first: No Server Connection needed on same subnet; enable Server-side connection only for remote access.
- Audio priority: 24–48 kbps Opus, 20 ms frames; target 80 ms end-to-end.
- Start video 1–3 Mbps H.264 (hardware).

Deploy & Ops

- Run as system services on all Pis
- Central logs & basic metrics (CPU, jitter, bitrate) for quick diagnostics.
- Backup signaling config and room ACLs with tokens/certifications regularly.

Rooms

Current Room:

test

2 / 10 participants (GUI sees: 2)

Leave Room

Debug Refresh

Available Rooms:

test (0/10)

Join Room

Create New Room:

+ Create Room

test



Start Call

2 participants

meltretras 1m ago
hello

You 1m ago
hello!

meltretras 1m ago
how is the project coming along

You now
:(

meltretras now
What happened

Connected Users (Server-wide) ×

Connected Users (Server-wide) ↻

All users connected to the server:
Currently showing: 2 users

iliketrains2314 (You)
In room: test
Online 1m
ID: 46bc9d84-c812-4983-a5dd-df4d89403d1f

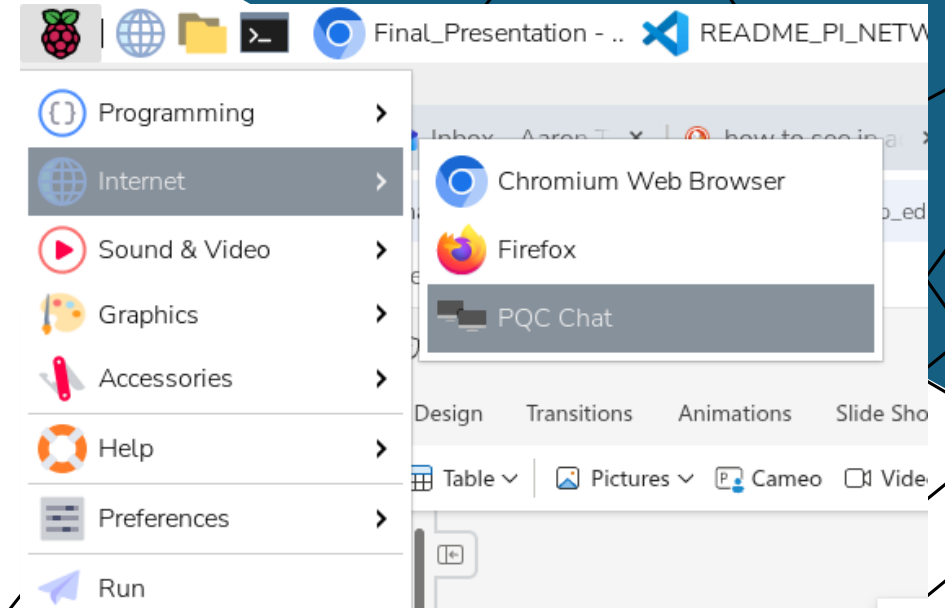
meltretras
In lobby
Online 1m
ID: 236b3be0-ec77-4b1e-80c6-3345151b3ca4

This project sucks :(

Send

Basic Setup

- Simple Implementation and Setup
 - Clone from git
 - Download Dependent Libraries
 - Run terminal commands
 - `cargo build --release`
 - `./scripts/deploy_pi_network.sh`
 - Open Gui from Pi's Menu
 - Type Address of server and Username to Connect (Default Address is server IP and Default Username is GitHub name)
 - Create/Join a Room and start Chatting/Talking!

A screenshot of the PQC Chat application interface. The title bar reads 'PQC Chat - Post-Quantum Secure Connection'. The interface has a dark theme. It displays the following fields: 'Server Host:' with the value '192.168.10.101', 'Port:' with the value '8443', and 'Username:' with the value 'iliketrains2314'. At the bottom, there is a 'Connect' button with a lock icon.

```
wl_registry#2 still attached
❖ iliketrains2314@raspberrypi:~/PQC-Post-Quantum-Chatting-12-6 $ cargo build --release
```

```
wl_registry#2 still attached
❖ iliketrains2314@raspberrypi:~/PQC-Post-Quantum-Chatting-12-6 $ ./scripts/deploy_pi_network.sh
```

Demo-Video

